

PORTFOLIO CASE STUDY

# ElbeFlow

## Hamburg Urban Mobility Lakehouse

Seven official urban layers, explainable analytics and a multilingual decision product

**84,191**

OFFICIAL STREAMS

**~459M**

SCHEDULED OBSERVATIONS

**102,994**

VERIFIED SAMPLE ROWS

**7**

OFFICIAL DATA LAYERS

PYTHON · PARQUET · DUCKDB · DBT · REACT · MAPLIBRE · TERRAFORM · OTEL

German by default · English · French · Arabic with right-to-left layout

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ILLUSTRATED PROJECT REPORT · AUGUST 2026

# A data platform that ends in a usable product

The project is designed to be evaluated at both engineering and product level.

**01****INGEST**

Paginated source adapters, bounded retries and incremental monthly windows.

**02****STORE**

Immutable JSONL/Gzip bronze and partitioned Parquet/ZSTD silver.

**03****MODEL**

DuckDB analytics, dbt transformations and explicit quality contracts.

**04****DELIVER**

Predictive, multilingual React product with cloud and observability assets.

## Honest scale accounting

The application separates exact counts from estimates. 84,191 streams is a catalogue count. 102,994 rows is the exact committed sample count. ~459 million is a conservative coverage-by-cadence estimate for scheduled sources only; traffic-light and charging events are excluded.

### Core reliability choices

- Idempotent partitions and half-open time windows
- Atomic completion, deduplication and SHA-256 verification
- Live API fallback without hiding freshness state
- Local-first design with a direct cloud migration path

# Executive dashboard

German is the default experience for the Hamburg target market.



Scale in seconds  
WHAT THE SCREEN PROVES

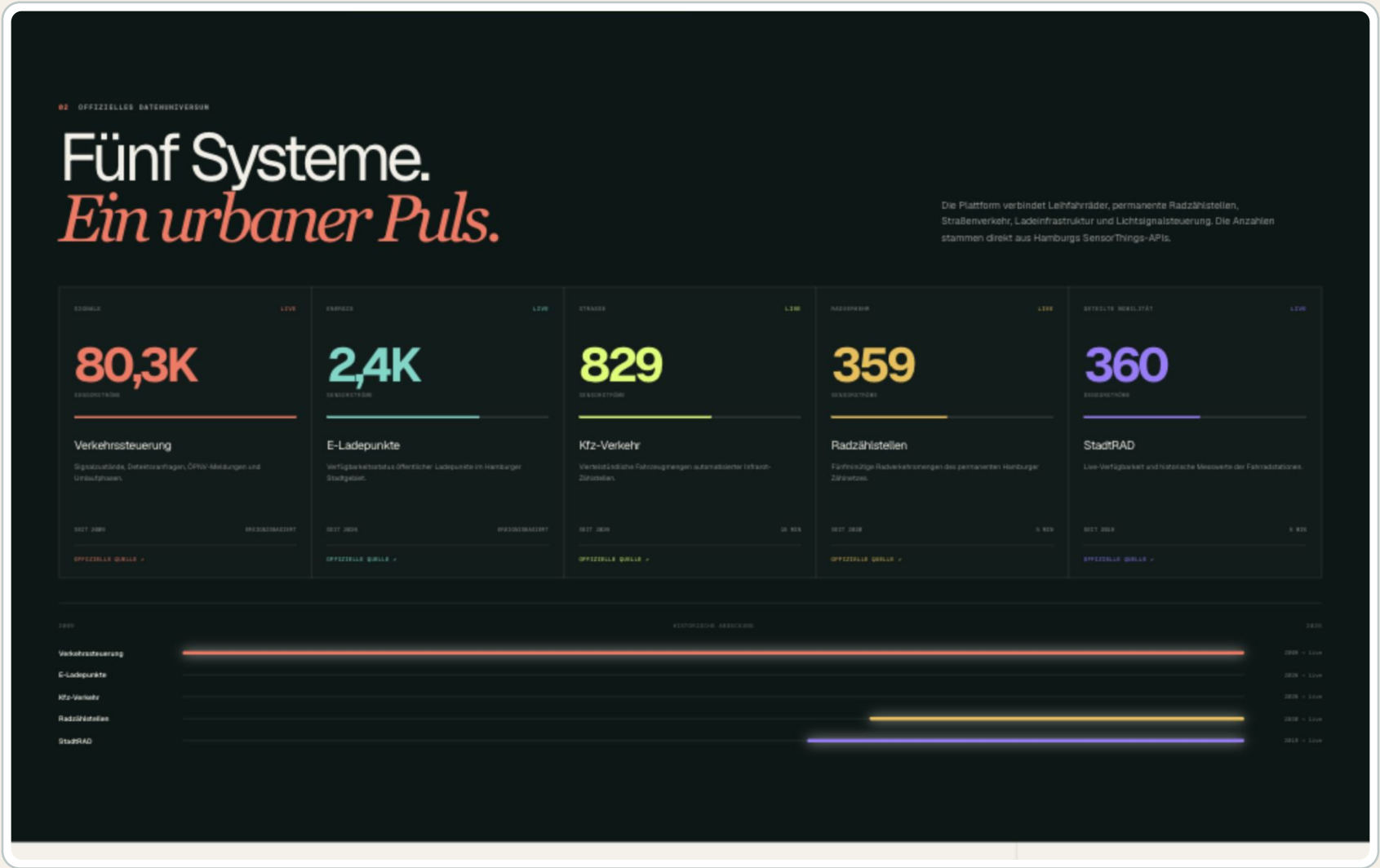
The opening view gives reviewers immediate evidence of the project's size, source diversity and reproducibility.

84,191 exact SensorThings streams • ~459 million scheduled records • 102,994 verified real sample rows • Seven official Hamburg data layers

Figure 1 — Default German overview with live-status indicator and headline metrics.

# Official source atlas

A single catalogue explains five very different operational datasets.



**Source diversity**  
WHAT THE SCREEN PROVES

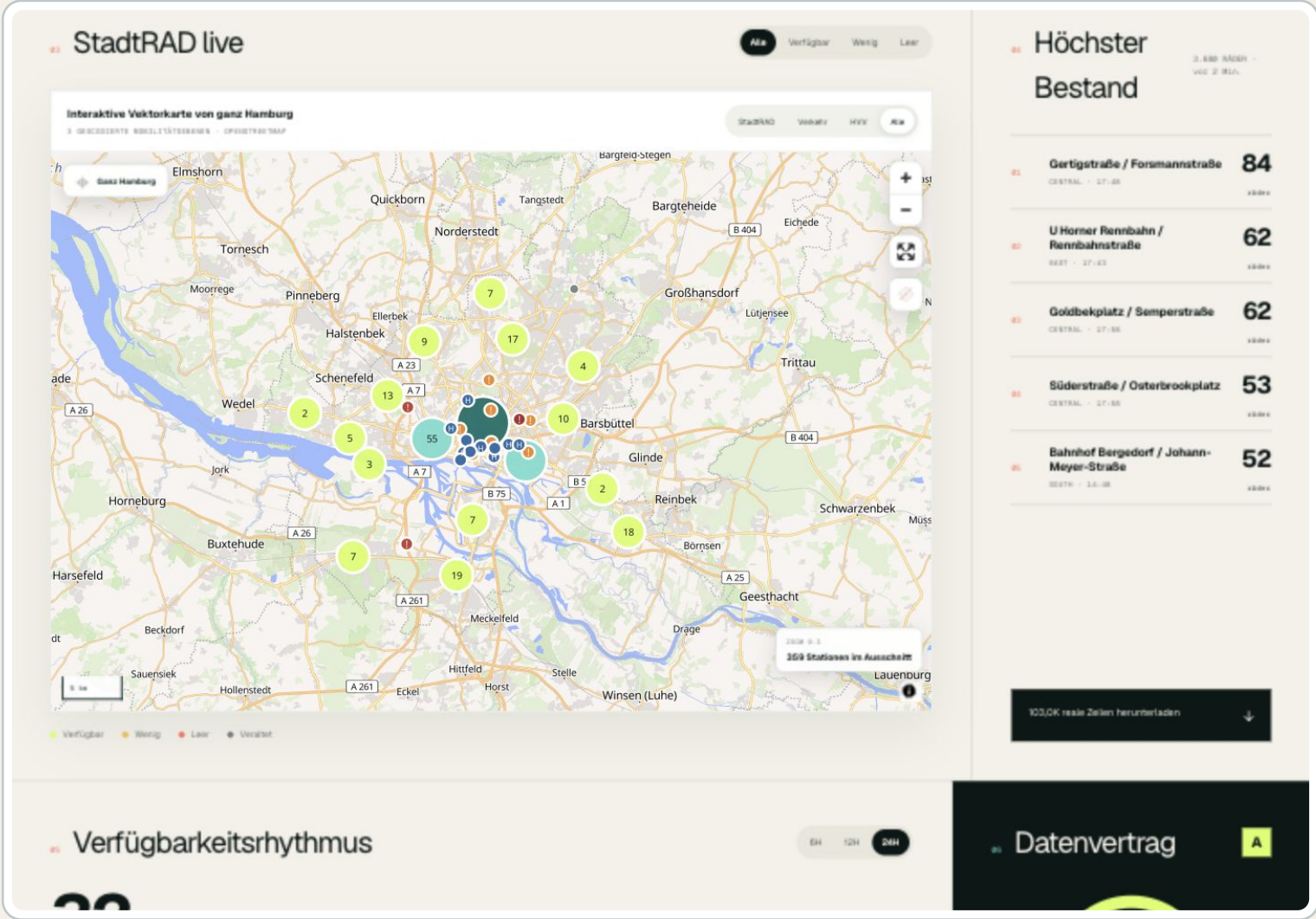
Each card communicates ownership, cadence, historical coverage, stream count and ingestion behavior.

Traffic-light detector events from 2009 • EV charging events and site status • 15-minute motor-traffic measurements • 5-minute bicycle counts and StadtRAD availability

Figure 2 — Source catalogue with data-domain metadata and coverage boundaries.

# Professional geographic city map

A MapLibre vector map turns official traffic, HVV and StadtRAD coordinates into a real spatial product.



**Coordinate integrity**  
WHAT THE SCREEN PROVES

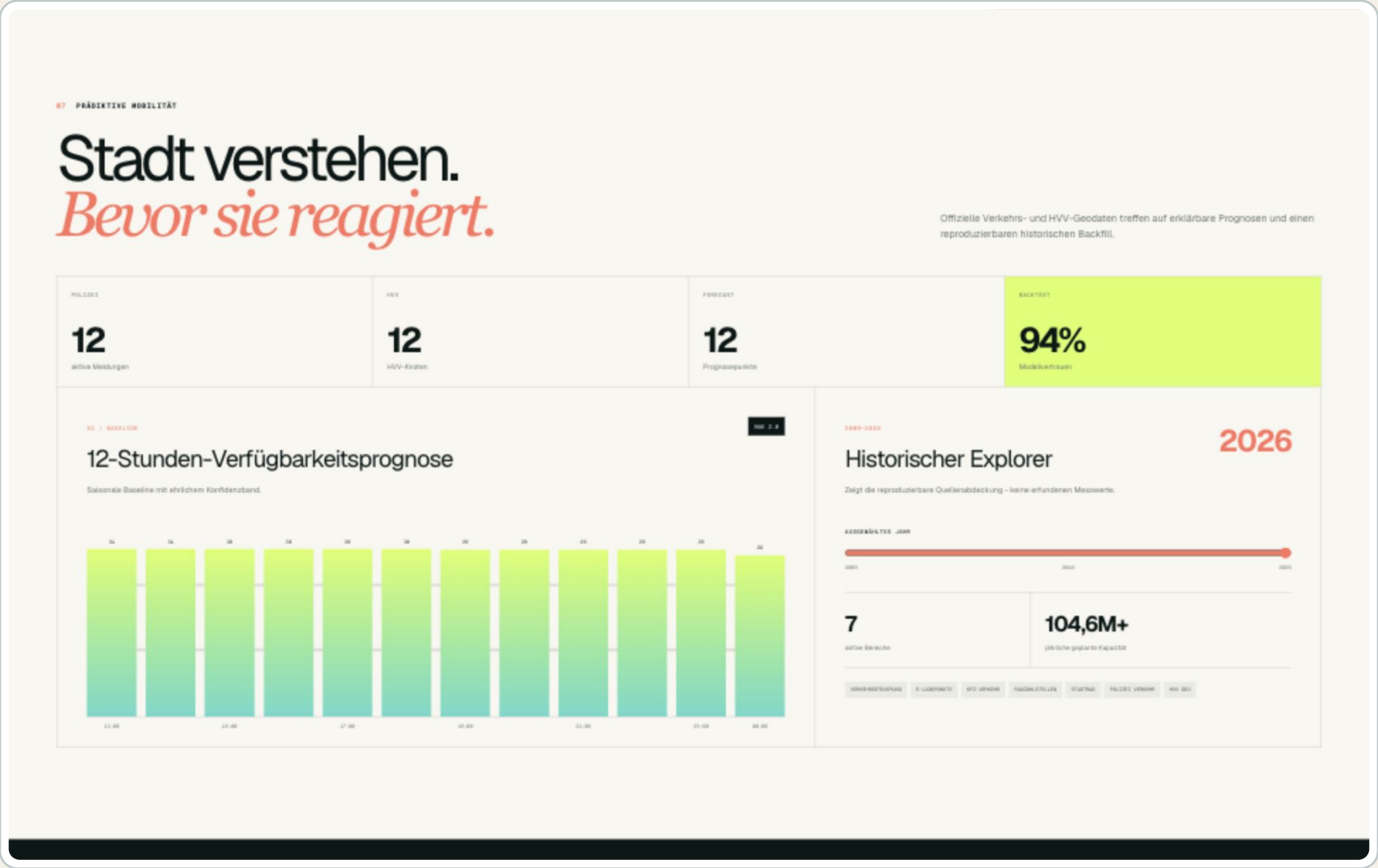
The map covers the full Hamburg operating area and keeps every point anchored to its official longitude and latitude while users pan, zoom and inspect layers.

12 current Polizei Hamburg traffic notices • 12 official HVV rail hubs and lines • 359 valid live station positions from the official feed • Dynamic clusters, zoom-dependent radii and street detail

Figure 3 — Full Hamburg extent on the MapLibre/OpenStreetMap vector basemap with all mobility layers enabled.

# Predictive mobility

Forecasts remain explainable, back-tested and explicit about uncertainty.



## No black-box claims

WHAT THE SCREEN PROVES

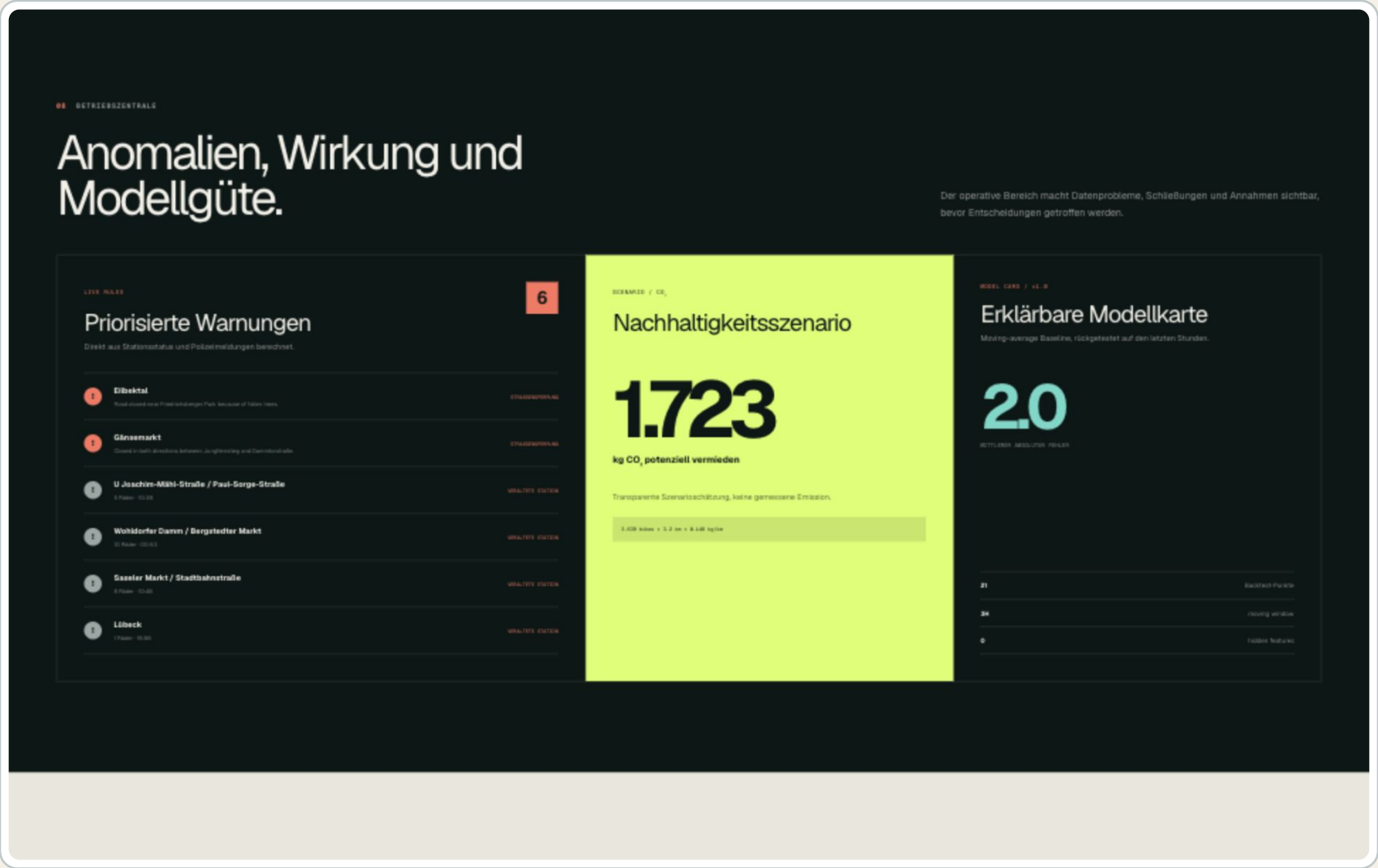
A rolling baseline produces a 12-hour availability outlook, while the historical explorer shows reproducible source coverage rather than invented observations.

12 forecast points with a confidence indicator • Mean absolute error from a rolling holdout • Interactive historical coverage from 2009 to 2026 • Scheduled capacity calculated from source cadence

Figure 4 — Explainable forecast KPIs and reproducible historical explorer.

# Operations and sustainability

Rule-based alerts, scenario impact and model quality share one decision surface.



## Operational honesty

WHAT THE SCREEN PROVES

Alerts come directly from station states and police closures. CO<sub>2</sub> is clearly labelled as a scenario estimate, never a measured emission.

Prioritised empty and stale station alerts • Current road closures from official WFS data • Transparent CO<sub>2</sub> formula and assumptions • Model card with MAE and holdout size

Figure 5 — Operations centre with anomaly, environmental and model-quality evidence.



# Quality and data contract

Reliability is presented as measurable product behavior.



## Trust is visible

WHAT THE SCREEN PROVES

Freshness, completeness, validity and uniqueness are shown beside the rules that protect downstream analytics.

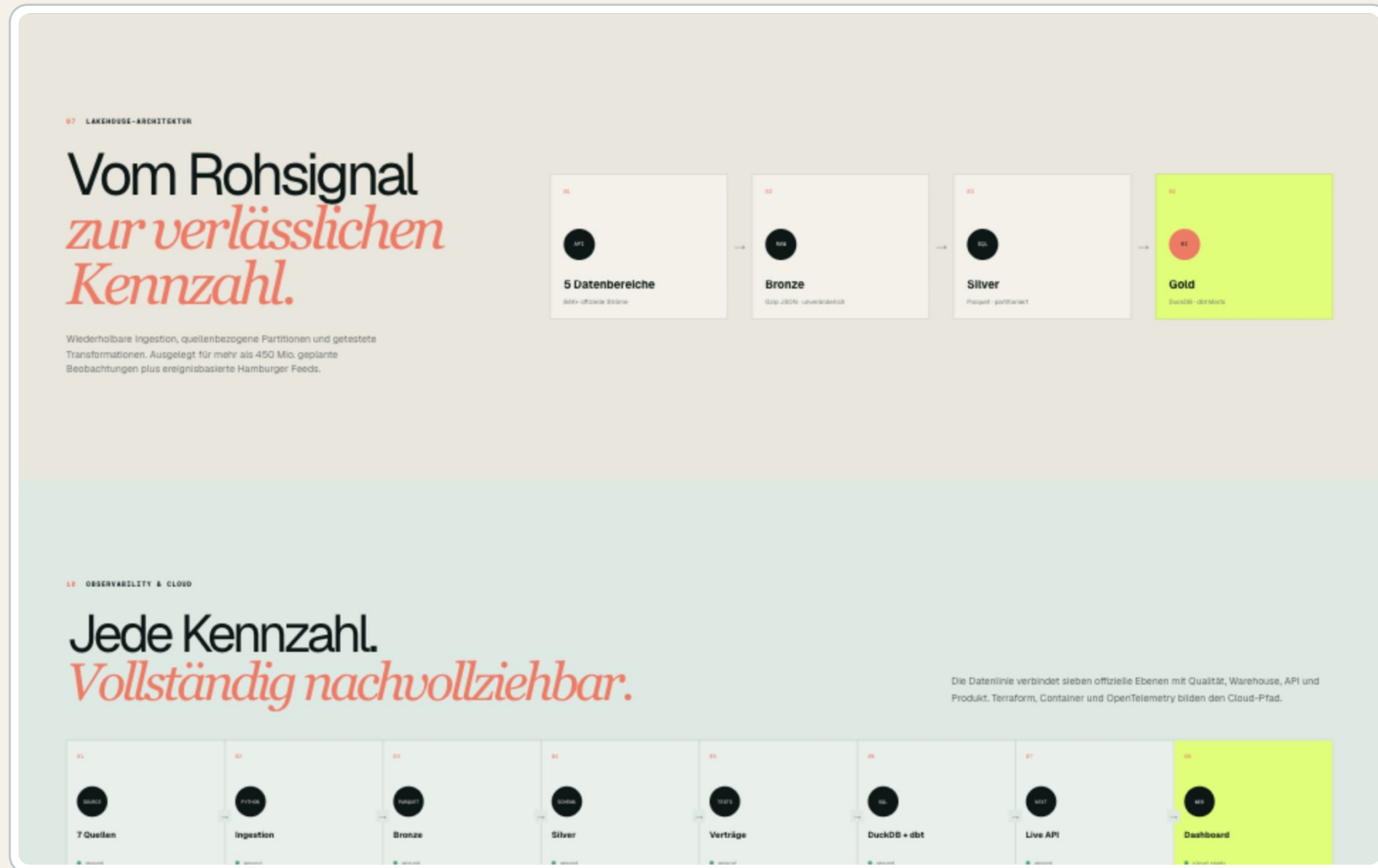
Required-key and timestamp validation • Duplicate observation protection • Domain checks for impossible negative counts • Clear separation of verified and estimated values

Figure 6 — Quality indicators and the explicit contract enforced by the pipeline.



# Lakehouse architecture

The dashboard explains how the repository turns APIs into analytics.



## End-to-end ownership

WHAT THE SCREEN PROVES

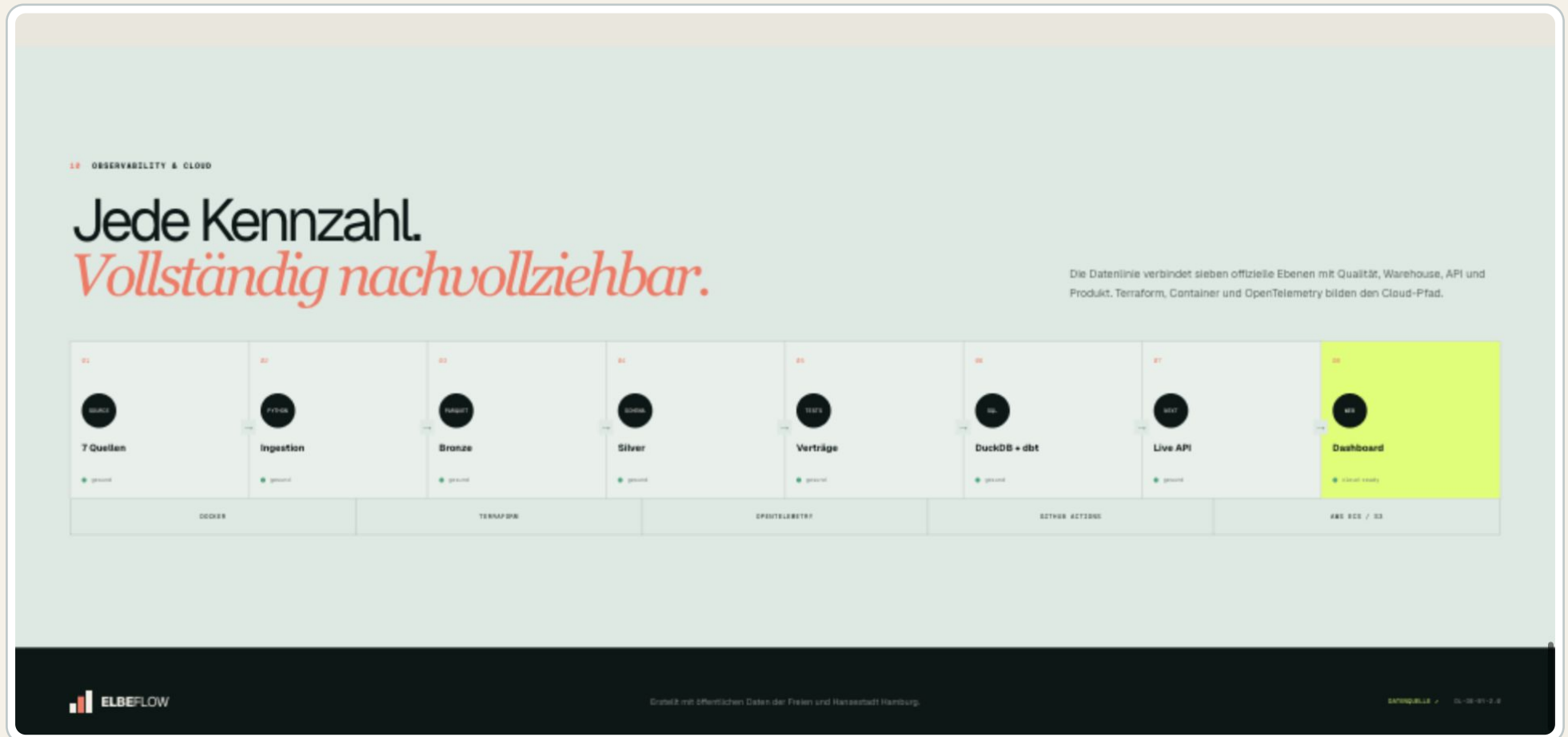
The architecture view connects ingestion, object storage, analytical modeling, orchestration and delivery.

*Immutable bronze for replayability • Partitioned Parquet silver for efficient scans • DuckDB + dbt for local analytics and contracts • Airflow orchestration and automated validation*

Figure 7 — Recruiter-readable bronze/silver/warehouse/product architecture.

# Lineage, observability and cloud

Every metric is connected from source to product and to a concrete deployment path.



### Production readiness

WHAT THE SCREEN PROVES

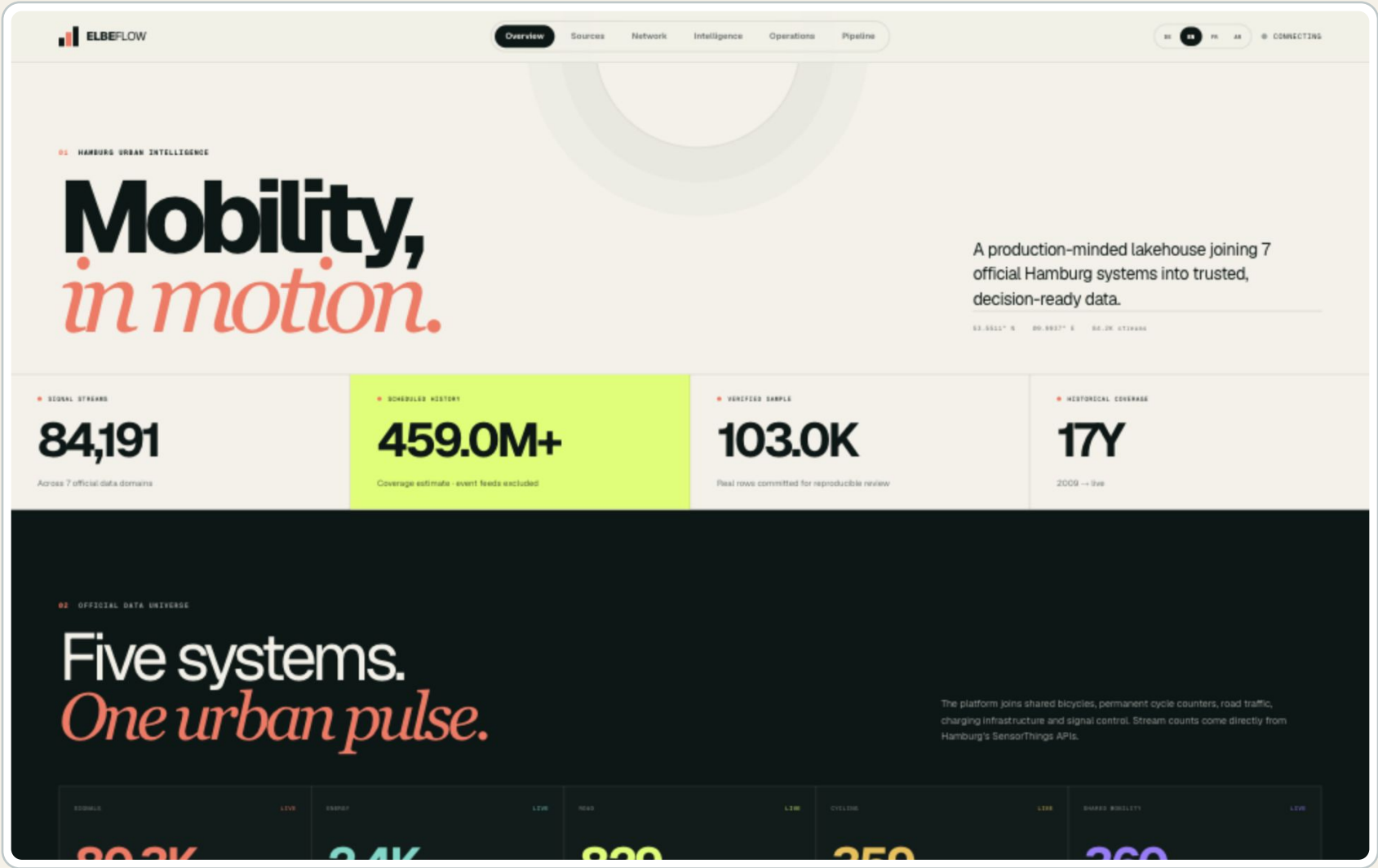
The visible lineage is backed by repository assets for containers, Terraform, OpenTelemetry and GitHub Actions validation.

Seven sources through bronze, silver and contracts • DuckDB + dbt analytical serving layer • Dockerised web and pipeline workloads • Terraform ECS/S3 reference and OTLP collector

Figure 8 — End-to-end lineage with health states and cloud-ready delivery assets.

# English interface

The full product can be evaluated by an international hiring team.



### International review

WHAT THE SCREEN PROVES

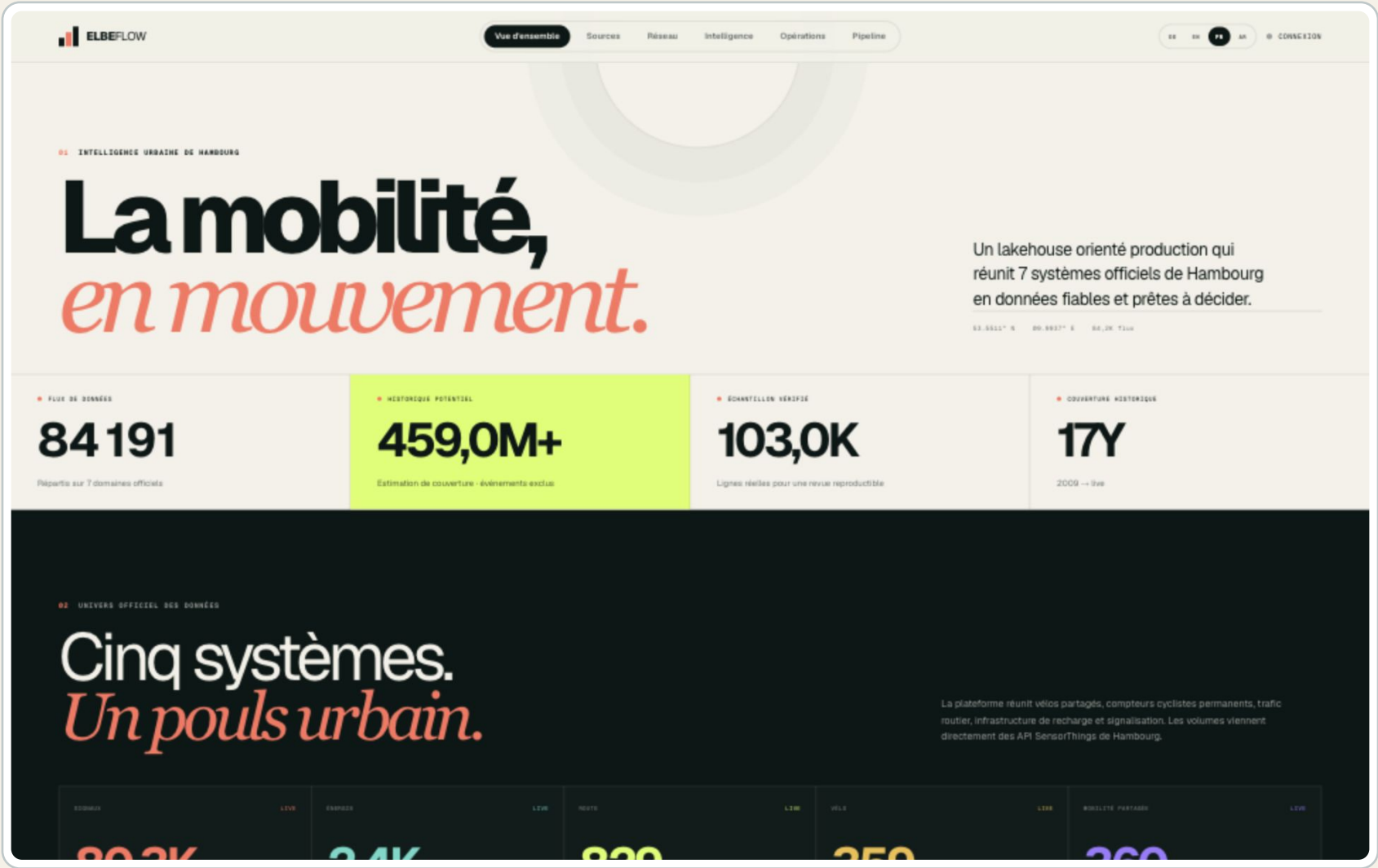
English localization covers navigation, source metadata, operational labels, quality explanations and formatting.

Complete UI localization • Locale-aware numeric and date formatting • Stable analytical meaning across languages • One-click language switching

Figure 9 — English overview preserving the same analytical hierarchy as German.

# French interface

Localization is implemented at the product layer, not as a static mock-up.



**Consistent semantics**  
WHAT THE SCREEN PROVES

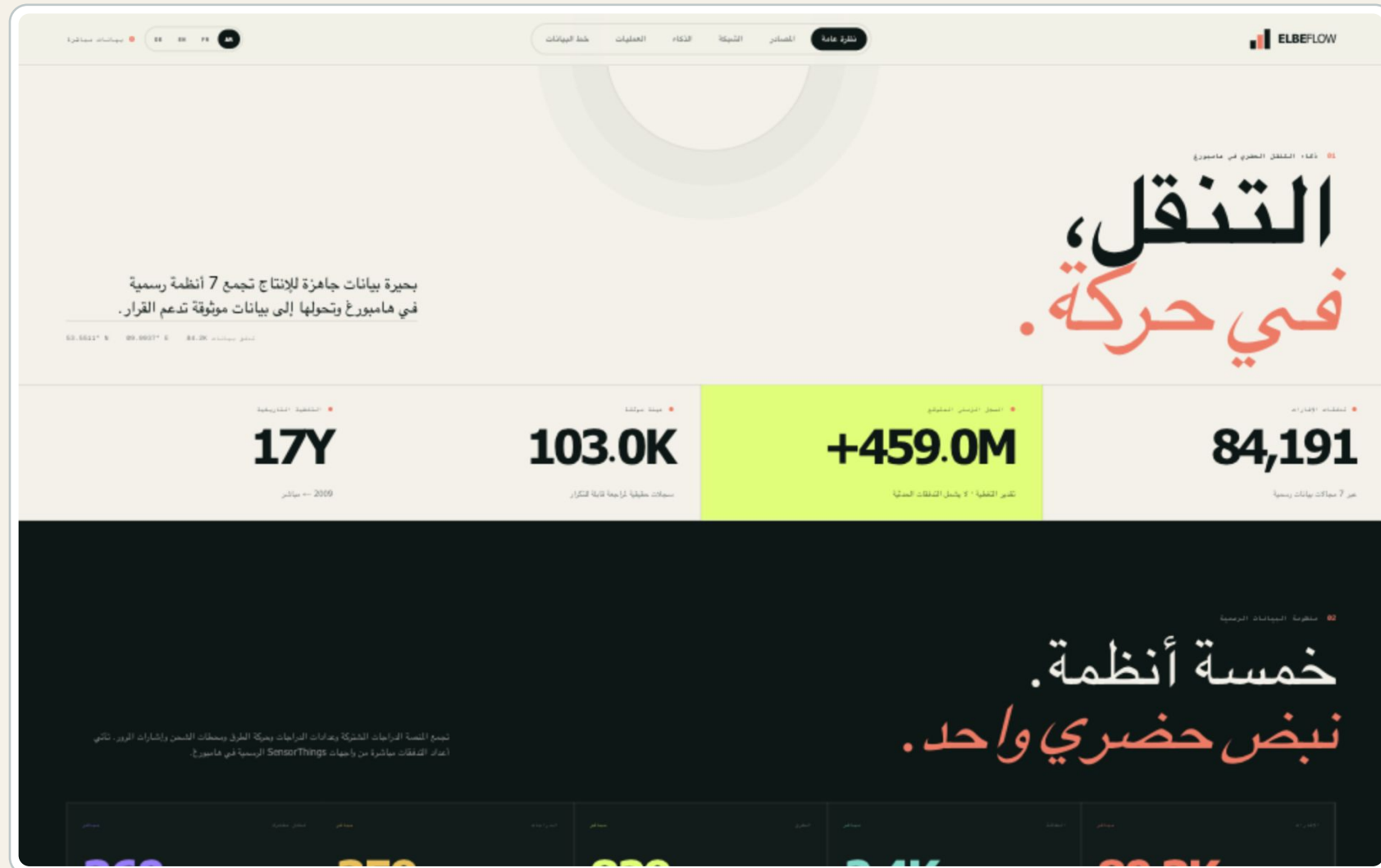
French users receive translated labels and explanations while every metric remains sourced from the same typed dataset.

*Translated product and engineering vocabulary • French number formatting • Identical data lineage and interactions • Shared component system avoids interface drift*

Figure 10 — French overview using the common data and component model.

# Arabic right-to-left interface

The application changes both language and document direction.



## Real RTL behavior

WHAT THE SCREEN PROVES

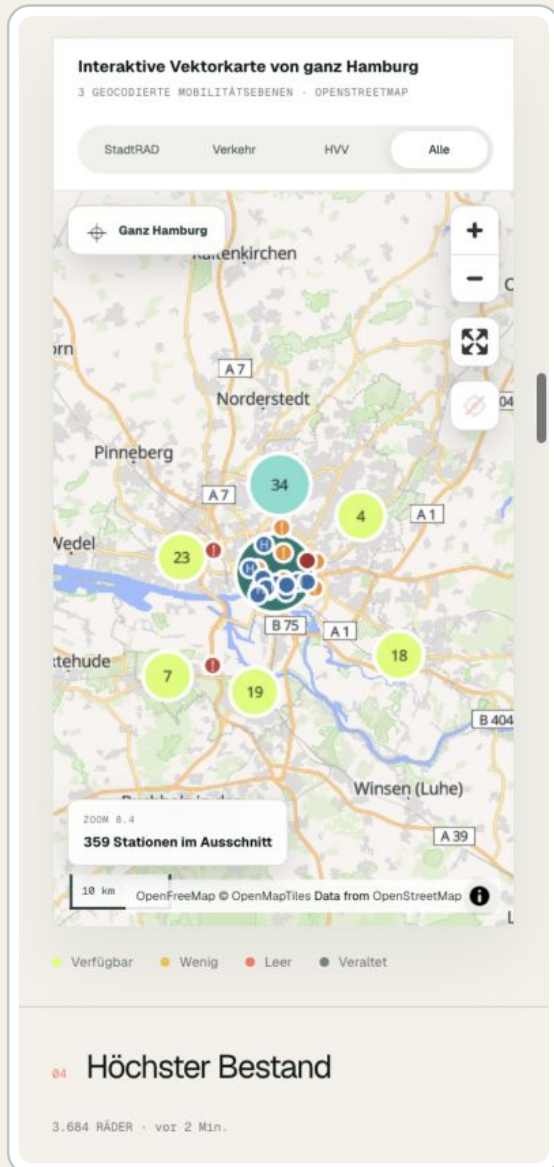
Arabic is not only translated: navigation, hierarchy, alignment and reading flow are mirrored for right-to-left use.

Document-level RTL direction • Arabic typography and translated interface copy • Layout-specific alignment corrections • Latin digits retained for technical readability

Figure 11 — Arabic overview with genuine RTL layout behavior.

# Mobile map experience

The same geographic product remains usable on a 390 × 844 viewport.



## RESPONSIVE PRODUCT

### A real map on every viewport

The mobile layout preserves the complete MapLibre interaction surface without horizontal overflow. Touch zoom, pan, clustering, layer controls and the full-Hamburg reset remain available in a readable single-column experience.

- Verified at 390 × 844 pixels
- No horizontal document overflow
- Touch zoom and pan stay enabled
- Map controls and layer filters remain reachable

Figure 12 — Full Hamburg vector map on the German mobile interface.